

Exploring Socio-Economic Trends in England and Wales: A Visual Analytics Approach

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1. Abstract

This project explores health inequalities and socio-economic disadvantages across Local Authority Districts (LADs) in England and Wales using 2011 and 2021 census data. The main objective is to examine whether areas with poorer health also show higher unemployment, lower education, poorer housing or any other forms of deprivation and how these patterns changed after the pandemic. The target users are public health analysts, local authority decision-makers and public health policy researchers who need to identify spatial patterns, compare local areas and understand where health disadvantage overlaps with socio-economic problems.

The data preparation process involved cleaning LAD level health and socio-economic activity datasets, creating comparable percentage and count measures and harmonizing the changing LAD geographies and codes between 2011 and 2021. This was necessary because the coursework guidance notes that geographic definitions do not map directly between census years. After harmonization, the final comparison dataset contained 331 comparable LADs in each year, giving 662 rows in total. Eight newer successor LADs were retained for analysis but excluded from filled-map views because Tableau's UK geocoding reference did not recognize the newer boundaries, this was treated as geocoding limitation rather than missing census data.

The final Tableau workbook contains four dashboards. The first dashboard provides a 2011 trends of poor health and related socio-economic data. The second compares harmonized 2011 and 2021 values, allowing users to switch between percentage and count views. The third and fourth dashboards are PCA and UMAP projections respectively, combined with Bayesian Gaussian Mixture Modelling to reveal clusters, outliers and uncertain LAD assignments. The visual analysis suggests that poor health is spatially uneven across LADs. As expected, it is mostly related to long-term sickness or disability, since both capture related aspect of health disadvantage. Positive relationships are also visible with low qualifications and unemployment while social rented housing shows a weaker relationship and private renting does not follow the same pattern. The PCA and UMAP dashboard reveal distinct LAD health and socio-economic profiles and BGMM confidence values highlight uncertain cases that require cautious interpretation.

2. Introduction

Health inequality is an important socio-economic issue because poor health is not distributed evenly across places and are influenced by wider social conditions. The Marmot Review argues that health is shaped not only by healthcare, but also by the conditions in which people are born, grow, live, work and age, including education, employment, income and housing [15]. Building on this idea, this project investigates whether Local Authority Districts(LADs) with poorer health outcomes also show signs of wider socio-economic disadvantage such as higher unemployment, lower education or poorer housing and how these patterns changes between 2011 and 2021 censuses.

The main objective of this project is to explore inequalities across England and Wales using visual analytics approach. The analysis uses LAD level data related to general health, economic activity, education and housing. The 2011 data provides a baseline while the 2021 data allows comparison that includes major social and economic across these two years[5],[6],[7],[8]. The comparison matters because the 2021 census was conducted during COVID-19 pandemic which gives a useful point for assessing whether health related disadvantage remained or shifted across different LADs.

The target users for this visualization are public health analysts, local authority decision makers and policy researchers. They need to identify areas with poor health and check if these regions overlaps with any other forms of disadvantages, compare all the local areas fairly and understand whether these inequalities are concentrated in particular parts of England and Wales. For example, a public health analyst may want to identify regions with bad health and compare them against unemployment or low qualification rates. Local authority teams may also need to distinguish between areas where many people are affected and areas where the proportion affected is unusually high.

The dashboards address five main questions:

1. Where are the LADs with highest levels of bad or very bad health?
2. Do areas with poorer health also show higher unemployment, lower education or poorer housing?
3. How did the health related disadvantage change between the years 2011 and 2021?
4. Which LADs appear as outliers and is different from the rest?
5. Can dimensionality reduction find groups of LADs with similar health and socio-economic profiles?

The aim is not only to answer the key questions but to give users a visual and intuitive way to explore health inequality across England and Wales. The visualization is structured as an analytical story across four dashboards. The first shows the 2011 baseline. The second compares 2011 and 2021 to highlight the changes over time. The third uses PCA to reveal the strongest linear structure in the socio-economic factors, as PCA reduces dimensionality while preserving as much as variation as possible [13]. The fourth uses UMAP to explore non-linear patterns, clusters and outliers that may not appear in the standard charts [12]. These projection views highlight patterns that may not be visible in maps or standard charts..

Overall, the project helps us understand where health inequality overlaps with socio-economic problems in England and Wales. By combining spatial maps, comparison charts and dimensionality reduction, the dashboard helps uncover this detailed LAD-level insights and examine how these relationships changed between 2011 and 2021.

3. Data Preparation and Abstraction

Data Extraction

This data preparation and abstraction is the most important part for loading the clean data into Tableau for visualization. The original datasets were downloaded from the “2011CensusIndexofTablesandTopics_v11_4_2(5)” . With over 1600 tables available only 4 relevant and clean tables were selected for the year 2011 namely general health, economic activity,

highest qualification and housing. The raw files were available as number, percentages and ranks. This project uses numbers and percentages, counts show the scale of affected population while percentages allows fair comparison between LADs with different population sizes.

For 2021, general health and economic activity data were extracted mainly as count-based datasets from Nomis Census 2021 sources [7], [8]. Percentage measures were then calculated during preprocessing. This follows the coursework guidance to use both absolute values and proportions when comparing areas of different population sizes.

Final Files	Original Census Table	Description
2011_general_health_LAD_clean	QS302EW	General health by LAD
2011_economic_activity_LAD_clean	QS601EW	Economic activity by LAD
2011_highest_qualification_LAD_clean	QS501EW	Highest qualification level by LAD
2011_housing_LAD_clean	QS405EW	Housing tenure by LAD
2021_general_health_Lad_clean	TS037	General health by LAD
2021_economic_activity_LAD_clean	TS066	Economic activity status by LAD

Data cleaning

The raw 2011 files required cleaning before they could be merged. In python, meta rows and the headers were removed and only LAD level rows were kept by filtering LAD code prefixes such as E06, E07, E08, E09 and W06 to ensure only England and Wales data were retained. Column names were changed to have consistent formats. Numeric values were cleaned by removing commas and converting text fields into numeric data types. This was important as Tableau and dimensionality process requires numerical columns.

Each topic in 2011 had two separate files for counts and percentage measures, they were cleaned and merged. Similar percentage measures were retained or calculated for unemployment, qualification levels and housing tenure. Percentages were required as the population vary across the LADs and counts show how many people were affected. The cleaned 2011 files had no missing values. After merging health, economic activity and housing data, the final dataset contained 348 LADs and 89 columns. A smaller 2011 health-economic dataset with 348 rows and 47 columns was also created for comparison with 2021.

For 2021 files, the cleaning involved renaming columns, selecting relevant fields, converting values to numeric format and calculating percentage measures because the dataset only had counts. The cleaned 2021 health and economic activity dataset contained 331 LADs and 41 columns with no missing values.

LAD Harmonization and Geographic Matching

LAD boundaries and codes changed between the 2011 and 2021 census. The coursework guidance notes that geographic definitions do not always match between 2011 and 2021 so harmonization was required before comparison [9]. A lookup table was created to map changed 2011 LAD codes to 2021 successor codes including merged LADs such as Bournemouth, Christchurch and Poole, Dorset, Buckinghamshire, East Suffolk and West Suffolk and code only changes such as Northumberland, Gateshead, St Albans, Stevenage and Welwyn Hatfield.

Wherever the old LADs had merged into newer authorities, the count variables were added first and then the percentages were calculated from newer totals rather than just using average to avoid misleading percentage values. After harmonization, the 2011 vs 2021 comparison dataset had 331 common LADs, producing 662 LAD records for the 2011 vs 2021 comparison.

After the census files were cleaned, mapping friendly columns such as LAD_Code and LAD_Name were added to final column list so that areas and codes could be recognized by Tableau geocoding package [10],[11]. The original fields area_code and area_name were not dropped for traceability as they show the original identifiers. Therefore, tableau geocoding package was used to map fields such as LAD_Code and LAD_Name. This was necessary because Tableau does not always recognize UK geographic boundaries automatically, and UK census mapping often needs additional geocoding support [10], [11]. The final 2011 dataset matched all 348 LADs to the Tableau reference.

Modelling and Projection Files

For dimensionality reduction percentage fields were selected rather than counts. Although counts could be standardized, they are still influenced by LAD population size. Percentages were more suitable because they allowed fairer comparison between LADs of different sizes.

For the 2011 projection, the modelling notebook used 39 percentage features across health, economic activity, education and housing. These variables were standardized using StandardScaler so that no variable dominated the projection because of scale differences [14]. PCA and UMAP were applied as two projection methods. PCA was used to obtain a linear projection while UMAP was used to explore non-linear patterns and local clusters [12],[13]. For 2011 PCA two components were tested first ,but six components were finalized based on cumulative explained variance. BGMM clustering was fitted using six-dimensional PCA while only PC1 and PC2 were displayed in Tableau. For UMAP, BGMM was fitted to two-dimensional UMAP. The exported projection files included LAD identifiers, projection coordinated, original percentage features, bgmm_cluster and bgmm_confidence fields.

The 2011 PCA output showed that PC1 and PC2 explained 53.4% of the variance, but the first six components explained 82.36%, so six PCA components were used for Bayesian Gaussian Mixture Model (BGMM) clustering, while only PC1 and PC2 were shown in Tableau for readability [4], [13], [14]. The BGMM produced cluster labels and confidence scores, which helped show LADs with similar profiles and highlight outliers. For 2011 vs 2021 comparison PCA, two components were kept because PC1 and PC2 explained 67.58% variation which was enough for a simple two-dimensional view [13].

For UMAP, two components were used for both 2011 and comparison datasets because aim was to create two -dimensional Tableau projection [12]. UMAP does not provide explained variance like PCA, so it was assessed through the visible grouping of LADs and the BGMM cluster outputs [12], [13]. In both PCA and UMAP , the maximum number of components for BGMM was fixed to six so that it gives the model enough flexibility to identify LAD profiles while keeping cluster views interpretable and confidence values showed how strongly each belonged to the assigned cluster [4],[14]

Data Abstraction

The final datasets were organized as tables, each row represented either one LAD or one LAD in specific year. The area fields such as `area_code`, `area_name`, `LAD_Code` and `LAD_Name`, were used to map each local authority in Tableau. The year field allowed comparison between 2011 and 2021. Count variables showed the number of people affected and percentage variables allowed fair comparison between LADs of different population sizes. The PCA and UMAP numbers shows where each LAD should appear in the projection charts. The `bgmm_cluster` and `bgmm_confidence` fields showed which cluster each LAD belonged to and how confident the model was about that choice. Organizing the data like this made it easier to use in Tableau because it clearly separated location and time information, census values and model results [1].

4. Task Definition

Using Munzner's task taxonomy, the dashboards were designed to support both exploratory and comparison based analysis [1]. The main goal is to understand where the poor health is concentrated across Local Authority Districts (LADs), whether it correlates to other socio-economic disadvantages and how did it change after the pandemic[5],[6],[7],[8].

The main task is discover, because the user are not searching for one specific value, they are exploring patterns, relationships and outliers [1]. Target users such as public analysts and local authority staff can browse the map to spot LADs with poor health and identify the exact health, unemployment, education and housing values for a selected area.

The second task is compare. Users compare different LADs against each other, compare counts with percentages and compare 2011 data with 2021 data to see whether health related disadvantages increased, reduced or shifted geographically [1].

The projection dashboards support summarise and explore tasks. PCA and UMAP helps with dimensionality reduction, helping the user to see the overall structure, clusters and outliers [12],[13]. Bayesian Gaussian Mixture Model(BGMM) cluster labels and confidence values help users interpret which LADs have similar profiles and how strongly they belong to a cluster.

In Munzner's what, why and how framework:

- What - LAD level census data including location fields, year, counts, percentages, PCA/UMAP coordinates, BGMM outputs.
- Why – to discover, identify, compare and summarize patterns in health inequality and socio-economic disadvantage.
- How – through, maps, bar charts, scatterplots, filters, tooltips and PCA/UMAP projection views.

Visualization Justification

Dashboard 1

Dashboard 1 shows the 2011 baseline for health inequality across LADs in England and Wales. The choropleth map is used because the first task is spatial and users need to see where bad health is concentrated. Color intensity shows the selected poor-health measure so that the users can easily identify broad regional patterns while keeping the geographic context of each LAD. This supports Munzner's browse, locate and identify tasks as user can see details for selected local authority [1]. The Top 15 bar chart supports this by ranking the worst-affected LADs more clearly, since bar

length is easier to compare than color differences on a map [1], [2]. The use of color intensity on the map supports spatial pattern recognition, but the bar chart is needed because human perception compares aligned lengths more accurately than subtle color differences [2], [3]. The map and Top 15 chart can switch between percentage and absolute count using the value selector. This was included because the two measures answer different user questions. Percentages allow fair comparison between LADs with different population sizes while counts shows the scale of affected residents. The Selected LAD Profile gives more detail when a user clicks an area. It shows that LAD's health, unemployment, education and housing values. Users can click on the map, the Top 15 chart or the scatter plot to update the profile, while the other views stay as overview context [1].

The scatterplot and selected LAD profile remain percentage based because their purpose is to compare socio-economic patterns fairly rather than reflect population size. The scatterplot was included because the dashboard needs to show relationships, not only locations and rankings. Each point represents one LAD with poor health percentage compared against a selected socio-economic factor. It also includes a linear trend line which summarizes the overall linear relationship between poor health and the selected socio-economic factor. This helps users to check whether LADs with higher poor health percentages also tend to have higher unemployment, lower qualification levels or housing disadvantage, while still keeping each LAD as a separate point so outliers can be easily identified. However, the line is interpreted as an exploratory visual summary rather than proof that one factor causes the other.

The dashboard also produced useful baseline findings from the 2011 Census data [5], [6]. In the percentage view, Merthyr Tydfil had the highest rate of bad or very bad health at 11.17% followed by Blaenau Gwent at 10.69% and Neath Port Talbot at 10.46%, showing that poor health was concentrated in parts of Wales and some post-industrial areas of northern England. The scatterplot shows that poor health is more strongly linked to some socio-economic factors than others. The strongest relationship is with long-term sick or disabled ($R^2 = 0.9282$), which is expected because it is closely related to health status. No qualifications also shows a strong positive relationship ($R^2 = 0.6719$), while unemployment shows a moderate one ($R^2 = 0.4143$). Social rented housing has a weaker link, and private renting shows almost none.

Dashboard 2

Dashboard 2 demonstrates how selected health and socio-economic measures changed between 2011 and 2021 in England and Wales. This dashboard is important because the project needs to compare change over time and not just describe the 2011 baseline. The choropleth map uses a diverging color scale to show whether the selected measure increased or decreased between the two years, where positive values show an increase and negative values show a decrease. This is suitable because change data has a meaningful midpoint at zero. The map supports Munzner's browse, locate and identify tasks by helping users see where change is concentrated and then inspect specific LADs in more detail [1]. The Top 15 bar chart supports the map by ranking the LADs with the largest increases which is easier to compare using bar length than color differences [1], [2], [3]. The value selector allows users to switch between percentage change and count measures.

The scatterplot was chosen because the dashboard needs to examine relationships between change measures and not only show where change happened. It is used to check whether changes in poor health move in the same direction as changes in the selected socio-economic measure. Each point

represents one LAD with the x-axis showing change in the socio-economic factor and the y-axis showing change in bad health. The trend line gives a visual summary of the relationship but it is only for exploration rather than proof that one factor causes the other.

The Selected LAD chart gives details by showing the LAD's 2011 and 2021 values side by side. Dashboard actions let users click on the map, Top 15 chart or scatterplot to update this view, while the other charts stay as overview context [1]. A hover highlight action is used in scatterplot to help users visually connect the same LAD across views without removing the wider pattern. The dashboard includes a separate chart for the eight LADs that appear as “unknown” on the choropleth map. These LADs were kept in the analysis, but Tableau could not recognize their new boundaries. Showing them separately avoids treating them as missing census data [9], [11].

Dashboard 2 shows that changes in poor health and changes in socio-economic conditions was not the same for every factor between 2011 and 2021 [5], [6], [7], [8]. The strongest relationship was for long term sick/disabled which showed a clear positive association with poor-health change ($R^2 = 0.6390$). A moderate relationship was observed for retired ($R^2 = 0.3808$) suggesting that LADs with larger increases in retired residents often saw increases in poor health change. In contrast, student inactive showed almost no relationship ($R^2 = 0.0045$) meaning not all types of economic inactivity were linked to health change.

Dashboard 3

Dashboard 3 uses Principal Component Analysis (PCA) to show LAD health and socio-economic features in a two-dimensional view. This dashboard was included because Dashboards 1 and 2 focus on single indicators while PCA helps to check whether LADs form broader groups based on many health and socio-economic variables at the same time. PCA was chosen because it is an interpretable linear dimensionality method that reduces many variables into a smaller number of components while keeping the main patterns in the data [13]. All percentage based features were standardized first so that no variable dominated the results due to differences in scale [14]. The scatterplot only shows PC1 and PC2 for readability but six PCA components were used for the Bayesian Gaussian Mixture Model (BGMM) because they retained more of the total variance in the data.

Each point in the PCA scatterplot represents one LAD. LADs that appear close together have more similar health and socio-economic profiles. Color shows the BGMM cluster label and point size shows bgmm_confidence which indicates how certain the model is about each assignment. BGMM was useful because it provides probabilities rather than forcing every LAD into a cluster with equal certainty [4]. The lowest confidence chart shows LADs that are most uncertain which help users identify boundary cases that may not clearly belong to one group. This supports Munzner's tasks of finding patterns, understanding groups and identifying outliers [1].

The profile charts provide details on demand. The cluster profile shows the average characteristics of each PCA/BGMM cluster while the selected LAD profile shows detailed values for one chosen LAD. Users can click a PCA point or an uncertain LAD to update the selected LAD profile supporting an overview to detail workflow [1]. Grouped bars were used for the cluster profile because the goal is to compare averages not add them together. This makes the chart interpretable and avoids the misleading impression that the values form a single total [2], [3].

Dashboard 3 shows that the PCA separated LADs into clear PCA/BGMM groups suggesting that some LADs have similar health and socio-economic profiles [13], [14]. In the 2011 view, Cluster 3 appears to have the high average bad health, unemployment and long term sickness/disability indicating a more disadvantaged profile. The lowest confidence chart shows that LADs such as Solihull, Ashford, Chorley, Bournemouth Christchurch and Poole, and South Derbyshire had low BGMM confidence values, around 0.35–0.39 meaning they are closer to cluster boundaries and should be interpreted more cautiously.

Dashboard 4

Dashboard 4 uses Uniform Manifold Approximation and Projection (UMAP) to show LAD health and socio-economic profiles in a two-dimensional non-linear view. This dashboard was included because Dashboard 3 uses PCA which is linear while UMAP can reveal local neighborhood patterns and non-linear groupings that PCA may not be able to detect [12],[13]. Each point represents one LAD and LADs that appear close together have more similar profiles.

Color shows BGMM cluster label and point size shows bgmm_confidence which indicates how certain the model is about each assignment. Color works well for cluster labels because it separates categories clearly while size gives a quick visual cue about stronger and weaker confidence values. BGMM was used because it gives probabilistic cluster membership rather than treating every LAD as equally certain [4], [14]. The Most uncertain UMAP/BGMM cluster assignments chart highlights LADs with the lowest confidence values helping users detect boundary cases that sit between clusters. This supports Munzner's tasks of finding patterns, summarizing groups and identifying outliers [1].

The profile charts give details on demand. The cluster profile shows the average characteristics of each UMAP/BGMM cluster while the selected LAD profile shows detailed values for one chosen LAD. Users can click a UMAP point or an uncertain LAD to update the selected LAD profile. Grouped bars are used because the goal is to compare averages and not add them together which avoids the misleading impression of a combined total [2], [3].

Dashboard 4 shows that UMAP reveals clearer local neighborhood groups than standard charts, suggesting that some LADs share similar combined health and socio-economic profiles [12]. In the 2011 view, clusters differ not only in poor health but also in unemployment, long-term sickness/disability, retirement and student inactivity. The uncertainty chart also shows that LADs such as Bexley, Broxbourne and Malvern Hills have lower confidence values (0.48–0.50) meaning their cluster assignments should be interpreted cautiously. Overall, Dashboard 4 complements PCA by showing non-linear structure, local grouping and uncertain boundary cases in the LAD profiles [1], [4], [12], [14].

Evaluation

The visualization was evaluated with three students from my discussion group. At the time of the evaluation only two dashboards were completed so the evaluation mainly focused on 2011 baseline dashboard and 2011 vs 2021 comparison dashboard. However, participants were told that PCA and UMAP dashboards were already planned and were asked whether projection views along with Bayesian gaussian mixture model would be useful for identifying clusters and outliers. Munzner's approach was taken into consideration, the evaluation checked whether dashboard supported the

intended analytical task and whether the interactions helped users move effectively from overview to detail [1].

Participants were asked to identify the LADs with poor health, select and LAD from Top 15 chart to interpret selected LAD and compare changed between the two years to check whether the dashboard was suitable for the target users. One useful question raised during feedback was who the dashboards were primarily designed for. This let me define the clear target audience as public health analysts, local authority decision makers and policy researchers. The main issue identified was that the first version only showed percentages which made it difficult to understand how many residents were affected in real numbers. To fix this, I added a percentage/count selector to the map and the Top 15 charts. The evaluation also showed that I needed clearer notes explaining which controls affected which charts and why some LADs appeared separately because of Tableau's geocoding limitations. Overall, the evaluation confirmed that the dashboards supported the main tasks and the feedback helped improve clarity and interpretation.

Conclusion

In summary, this project analyzed health inequality and socio-economic disadvantage across Local Authority Districts in England and Wales and how these patterns changed across the years 2011 and 2021. The results showed that poor health was not evenly distributed across the nation. In the 2011 percentage view, Merthyr Tydfil had the highest bad or very bad health rate at 11.17% followed by Blaenau Gwent at 10.69% and Neath Port Talbot at 10.46% ,showing concentration of poor health in parts of Wales and post-industrial areas [5], [6]. The count view gave a different answer where larger authorities such as Birmingham had the highest absolute number of affected residents with 71,791 people reporting bad or very bad health. This confirmed why both percentage and count views were needed. The scatterplots showed that poor health was more strongly related to some socio-economic indicators than others. In 2011, long-term sickness/disability had the strongest relationship with poor health followed by no qualifications and unemployment, while private renting showed little relationship. In the 2011–2021 comparison dashboard, change in long-term sickness/disability also had the clearest association with poor-health change, while student inactivity showed almost none. This suggests that health inequality is linked to wider socio-economic disadvantage but not every indicator is equally related to health [7], [8], [15], [16]. PCA showed broader linear groupings of LADs while UMAP revealed clearer local neighborhood groups with similar profiles [12], [13]. BGMM cluster labels and confidence values helped identify which LADs clearly belonged to a cluster and which were closer to boundaries requiring more cautious interpretation [4], [14]. Overall, the findings show that health inequality is part of a wider pattern of education, employment, inactivity and population structure rather than a health measure alone.

This coursework showed that effective visualization depends on choosing chart types that match the user's task. Choropleth maps supported spatial overview, Top 15 bar charts enabled clear ranking, scatterplots helped examine relationships, and selected LAD profiles provided details on demand. PCA and UMAP projections were especially useful for showing multivariate clusters and outliers that were not visible from single charts alone [1], [12], [13]. A key design lesson was that the same dataset can answer different questions depending on how it is encoded. Percentages supported fair comparison between LADs while counts showed the scale of affected residents. User feedback also highlighted the importance of clear labels, tooltips and notes, especially for explaining which controls affected which views and why some LADs were shown separately

because Tableau did not recognize newer boundaries. Overall, the project showed that information visualization is not just about making dashboards look attractive but about designing clear, interactive and task focused views that help users interpret complex census data accurately [1], [2], [3].

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